WoSM Academy

MAT231-F27 · CALCULUS II

# Convergence tests

## Comparison, integral, ratio and root tests

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WoSM Academy

Lecture 02 · Unit 2 · Fall 2027



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# Last time

- $\sum a_k$ converges if its partial sums $S_n = a_1 + \cdots + a_n$ converge.
- Geometric series: $\sum_{k=0}^{\infty} r^k = \frac{1}{1-r}$ if $|r| < 1$; divergent if $|r| \geq 1$.
- $n$-th term test: if $\sum a_k$ converges, then $a_k \to 0$ — not conversely.
- Telescoping series: $\sum (b_k - b_{k+1}) = b_1 - \lim b_n$.
- $\sum \frac{1}{k} = \infty$, but $\sum \frac{1}{k^2} \leq 2$: its partial sums are bounded.

For most series we cannot compute $S_n$ exactly. Today: how to decide *whether* a series converges without finding its sum.

# This lecture

01 Series with nonnegative terms

02 The integral test

03 Comparison tests

04 The ratio test

05 The root test

06 Choosing a test

07 Summary and practice

Reading: G. Strang, E. Herman, *Calculus, Volume 2* (OpenStax), Sections 5.3, 5.4 and 5.6.

01

# Series with nonnegative terms

# Bounded partial sums

**Theorem**

If $a_k \geq 0$ for all $k$, then $(S_n)$ is increasing, and $\sum a_k$ converges if and only if $(S_n)$ is bounded above. Otherwise $\sum a_k = \infty$.

Proof: $S_{n+1} = S_n + a_{n+1} \geq S_n$; use the monotone convergence theorem.

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- Today's tests assume $a_k \geq 0$ (for $k \geq N$ is enough). Both signs: Lecture 3.

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**Remark**

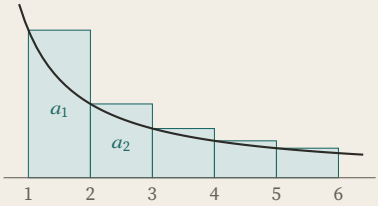
Tests decide *whether* a series converges, rarely what the sum is: no simple closed form is known for $\sum_{k=1}^{\infty} \frac{1}{k^3}$.

02

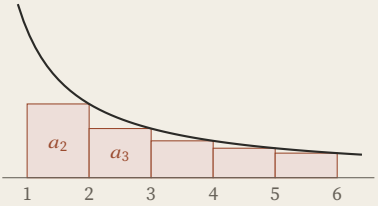
# The integral test

# Sums and integrals

Let $f$ be positive and decreasing, $a_k = f(k)$. Rectangles of width 1 against the area under the graph (Calculus I, Unit 26):




The rectangle on $[k, k+1]$ has height $a_k$ and lies *above* the graph, so $\int_1^{n+1} f(x)\,\mathrm{d}x \le a_1 + \cdots + a_n$.




The rectangle on $[k-1, k]$ has height $a_k$ and lies *below* the graph, so $a_2 + \cdots + a_n \le \int_1^n f(x)\,\mathrm{d}x$.

# The integral test

**Theorem (Integral test)**

Let $f\colon [1,\infty) \to [0,\infty)$ be decreasing, and $a_k = f(k)$. Then $\sum_{k=1}^{\infty} a_k$ converges if and only if $\int_1^{\infty} f(x)\,\mathrm{d}x$ converges, and

$$\int_1^{n+1} f(x)\,\mathrm{d}x \le S_n \le a_1 + \int_1^{n} f(x)\,\mathrm{d}x \qquad \text{for every } n.$$

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**Proof.**

$a_{k+1} \le \int_k^{k+1} f \le a_k$ as $f$ decreases; sum over $k$. $F(t) = \int_1^t f$ increases, so $\int_1^{\infty} f$ converges iff $F$ is bounded, iff $(S_n)$ is bounded. □

# $p$-series

**Theorem ($p$-series)**

$\sum_{k=1}^{\infty} \frac{1}{k^p}$ converges if and only if $p > 1$.

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**Proof.**

If $p \le 0$, the terms do not tend to 0. If $p > 0$, then $f(x) = x^{-p}$ is positive and decreasing, and $\int_1^t x^{-p}\,dx$ equals $\frac{t^{1-p}-1}{1-p}$ for $p \neq 1$ and $\ln t$ for $p = 1$. It has a finite limit as $t \to \infty$ if and only if $p > 1$ (Calculus I, Unit 23). □

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So $\sum \frac{1}{\sqrt{k}}$ and $\sum \frac{1}{k}$ diverge, while $\sum \frac{1}{k^{1.01}}$ and $\sum \frac{1}{k^2}$ converge. These are the standard series to compare with.

# Logarithms at the boundary

The test works on $[2,\infty)$ too. Both functions below are positive and decreasing there; substitute $u=\ln x$.

- $\displaystyle\sum_{k=2}^{\infty}\frac{1}{k\ln k}=\infty,$ since $\displaystyle\int_2^t\frac{\mathrm{d}x}{x\ln x}=\ln\ln t-\ln\ln 2\to\infty.$
- $\displaystyle\sum_{k=2}^{\infty}\frac{1}{k(\ln k)^2}<\infty,$ since $\displaystyle\int_2^t\frac{\mathrm{d}x}{x(\ln x)^2}=\frac{1}{\ln 2}-\frac{1}{\ln t}\to\frac{1}{\ln 2}.$

# Logarithms at the boundary

The test works on $[2, \infty)$ too. Both functions below are positive and decreasing there; substitute $u = \ln x$.

- $\displaystyle\sum_{k=2}^{\infty} \frac{1}{k \ln k} = \infty,$ since $\displaystyle\int_2^t \frac{\mathrm{d}x}{x \ln x} = \ln\ln t - \ln\ln 2 \to \infty.$
- $\displaystyle\sum_{k=2}^{\infty} \frac{1}{k (\ln k)^2} < \infty,$ since $\displaystyle\int_2^t \frac{\mathrm{d}x}{x (\ln x)^2} = \frac{1}{\ln 2} - \frac{1}{\ln t} \to \frac{1}{\ln 2}.$

How slowly does the first one diverge? The upper bound of the integral test, started at $k = 2$:

$$\sum_{k=2}^{n} \frac{1}{k \ln k} \;\le\; \frac{1}{2 \ln 2} + \ln\ln n - \ln\ln 2.$$

For $n = 10^{100}$ the right-hand side is about $6.53$. The series diverges, but no computer will ever show it.

# How good is a partial sum?

**Theorem (Remainder estimate)**

Under the conditions of the integral test, if $\sum a_k = S$ converges, then the remainder $R_n = S - S_n = \sum_{k=n+1}^{\infty} a_k$ satisfies $\int_{n+1}^{\infty} f(x)\,\mathrm{d}x \le R_n \le \int_n^{\infty} f(x)\,\mathrm{d}x$.

Proof: the same rectangles, starting at $n+1$ instead of 1.

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Proof: the same rectangles, starting at $n+1$ instead of 1.

> **Example ($\sum \frac{1}{k^2}$ with $n = 10$)**
>
> $S_{10} = 1.549768\ldots$ and $\frac{1}{11} \le R_{10} \le \frac{1}{10}$, so $1.64068 \le S \le 1.64977$. The true value is $\frac{\pi^2}{6} = 1.644934\ldots$; $S_{10}$ alone is off by almost 0.1.

# Two warnings

### The integral is not the sum

$\int_1^\infty \frac{\mathrm{d}x}{x^2} = 1$, but $\sum_{k=1}^\infty \frac{1}{k^2} = \frac{\pi^2}{6}$. The test says *whether* the series converges; the bounds say roughly *where*.

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### The function must be decreasing

$f(x) = \sin^2(\pi x)$ is nonnegative, and $a_k = f(k) = 0$ for every $k$, so $\sum a_k = 0$ converges. But $\int_1^\infty \sin^2(\pi x)\,\mathrm{d}x = \infty$. Without monotonicity the rectangles say nothing about the graph between the integers.

03

# Comparison tests

# The comparison test

**Theorem (Comparison test)**

Suppose $0 \leq a_k \leq b_k$ for all $k \geq N$.

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2. If $\sum a_k$ diverges, then $\sum b_k$ diverges.

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**Proof.**

Only the tail matters, so take $N = 1$. Then $\sum_{k=1}^{n} a_k \le \sum_{k=1}^{n} b_k \le \sum_{k=1}^{\infty} b_k$: the partial sums of $\sum a_k$ increase and are bounded, so they converge. Part 2 is the contrapositive of part 1. □

In short: below a convergent series converges; above a divergent series diverges.

# Comparison: examples

- $\sum \dfrac{1}{2^k + k}$
- $\sum \dfrac{\sin^2 k}{k^2}$
- $\sum \dfrac{1}{k!}$
- $\sum \dfrac{\ln k}{k}$
- $\sum \dfrac{1}{\sqrt{k^2 + 1}}$

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- $\sum \dfrac{1}{\sqrt{k^2 + 1}}$ : $\sqrt{k^2 + 1} \le \sqrt{2}\, k$, so the term is $\ge \dfrac{1}{\sqrt{2}\, k}$. Diverges.

# Compare in the right direction

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> $\frac{1}{k^2+1} \le \frac{1}{k}$ is true but useless: being below a *divergent* series proves nothing. Neither does being above a convergent one.

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# The limit comparison test

**Theorem (Limit comparison test)**

Let $a_k \geq 0$, $b_k > 0$ and $\dfrac{a_k}{b_k} \to L$.

1. If $0 < L < \infty$, then $\sum a_k$ and $\sum b_k$ both converge or both diverge.
2. If $L = 0$ and $\sum b_k$ converges, then $\sum a_k$ converges.
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**Proof.**

1. For $k \geq N$: $\frac{L}{2} \leq \frac{a_k}{b_k} \leq 2L$, so $\frac{L}{2} b_k \leq a_k \leq 2L\, b_k$; compare both ways. 2 and 3: eventually $a_k \leq b_k$, respectively $a_k \geq b_k$. □

# Choosing what to compare with

Keep the dominant terms; compare with $\frac{1}{k^p}$ or $r^k$.

- $\displaystyle\sum_{k=3}^{\infty} \frac{1}{k^2-5}$:
- $\displaystyle\sum \frac{3k^2+7k}{k^4+2}$:
- $\displaystyle\sum \frac{1}{\sqrt{k^3+1}}$:
- $\displaystyle\sum \frac{2k+1}{k^2+\sqrt{k}}$:
- $\displaystyle\sum \sin\frac{1}{k}$:
- $\displaystyle\sum \left(1-\cos\frac{1}{k}\right)$:

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- $\displaystyle\sum \frac{1}{\sqrt{k^3+1}}$: $b_k = \frac{1}{k^{3/2}}$, $L = 1$, $p = \frac{3}{2} > 1$. Converges.
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# Comparison in $O$-notation

For $a_k, b_k \geq 0$:

- If $a_k = O(b_k)$ and $\sum b_k$ converges, then $\sum a_k$ converges $(a_k \leq C b_k$ for $k \geq N)$.
- If $a_k = \Theta(b_k)$, the two series converge or diverge together.

So convergence depends only on how fast the terms decrease. From slow to fast:

$$\underbrace{\frac{1}{\sqrt{k}},\ \frac{1}{k},\ \frac{1}{k \ln k}}_{\text{diverge}} \qquad \underbrace{\frac{1}{k(\ln k)^2},\ \frac{1}{k^{1.01}},\ \frac{\ln k}{k^2},\ \frac{1}{k^2},\ \frac{k^{10}}{2^k},\ \frac{1}{k!}}_{\text{converge}}$$

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> **Remark**
>
> $\ln k = O(k^{\varepsilon})$ for every $\varepsilon > 0$, so $\sum \frac{(\ln k)^m}{k^p}$ converges for every $m$ when $p > 1$.

04

# The ratio test

# The ratio test

In a geometric series the ratio $\frac{a_{k+1}}{a_k} = r$ is constant. If the ratio is eventually below some $q < 1$, the series does at least as well as a geometric series with ratio $q$.

**Theorem (Ratio test)**

Let $a_k > 0$ and $\dfrac{a_{k+1}}{a_k} \to L$, where $L$ may be $\infty$.

1. If $L < 1$, then $\sum a_k$ converges.
2. If $L > 1$, then $\sum a_k$ diverges.
3. If $L = 1$, the test gives no information.

# Proof of the ratio test

**Proof.**

1. Choose $q$ with $L < q < 1$. There is an $N$ with $a_{k+1} \le q\, a_k$ for all $k \ge N$, so by induction
$$a_k \le a_N\, q^{k-N} \qquad (k \ge N).$$
$\sum_k a_N q^{k-N}$ is a geometric series with ratio $q < 1$; compare.
2. There is an $N$ with $a_{k+1} > a_k$ for all $k \ge N$. The terms are positive and increasing, so they do not tend to 0: the series diverges.
3. For $a_k = \frac{1}{k}$ and for $a_k = \frac{1}{k^2}$ the ratio tends to 1. The first series diverges, the second converges. □

# Ratio test: examples

- $\sum \frac{k}{2^k}$:
- $\sum \frac{2^k}{k!}$:
- $\sum \frac{3^k}{k^3}$:
- $\sum \frac{(2k)!}{(k!)^2\, 5^k}$:
- $\sum \frac{k!}{k^k}$:

# Ratio test: examples

- $\sum \dfrac{k}{2^k}$: $\dfrac{k+1}{2^{k+1}} \cdot \dfrac{2^k}{k} = \dfrac{k+1}{2k} \to \dfrac{1}{2}$. Converges.
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- $\sum \dfrac{3^k}{k^3}$: $3\left(\dfrac{k}{k+1}\right)^3 \to 3$. Diverges.
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- $\sum \dfrac{k!}{k^k}$: $\dfrac{(k+1)!}{(k+1)^{k+1}} \cdot \dfrac{k^k}{k!} = \left(\dfrac{k}{k+1}\right)^k = \dfrac{1}{(1+\frac{1}{k})^k} \to \dfrac{1}{e}$. Converges.

# Ratio test: examples

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The ratio test suits factorials and exponentials: most of the factors cancel.

# Exponentials beat powers

**Example** ($\sum \frac{k^{100}}{1.01^k}$)

$$\frac{a_{k+1}}{a_k} = \left(1+\frac{1}{k}\right)^{100} \cdot \frac{1}{1.01} \;\to\; \frac{1}{1.01} < 1.$$

The series converges.

# Exponentials beat powers

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The series converges.

- Yet the terms *grow* until $k \approx 10\,050$, where $a_k \approx 10^{356}$, and fall below 1 only after $k \approx 117\,000$. Convergence is decided by the tail; any finite number of terms is irrelevant.
- The same argument: $\sum k^m r^k$ converges for every power $m$ and every $0 < r < 1$. In particular $k^m r^k \to 0$.

# When the ratio test fails

**$L = 1$**

For every series whose terms are powers of $k$, or quotients of polynomials in $k$, the ratio tends to 1:

$$\frac{1/(k+1)^p}{1/k^p} = \left(\frac{k}{k+1}\right)^p \to 1.$$

The ratio test says nothing about them. Use a comparison or the integral test.

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### No limit

The test needs the limit of $\frac{a_{k+1}}{a_k}$ to exist. If the ratios oscillate, try the root test (next section).

05

# The root test

# The root test

**Theorem (Root test)**

Let $a_k \geq 0$ and $\sqrt[k]{a_k} \to L$, where $L$ may be $\infty$. If $L < 1$, then $\sum a_k$ converges. If $L > 1$, it diverges. If $L = 1$, the test gives no information.

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**Proof.**

- $L < 1$: choose $L < q < 1$. For $k \geq N$, $\sqrt[k]{a_k} \leq q$, that is $a_k \leq q^k$. Compare with the geometric series $\sum q^k$.
- $L > 1$: for $k \geq N$, $\sqrt[k]{a_k} \geq 1$, so $a_k \geq 1$ and $a_k \not\to 0$.
- $L = 1$: again $\frac{1}{k}$ and $\frac{1}{k^2}$, because $\sqrt[k]{k} \to 1$. □

# Root test: examples

Useful limits: $\sqrt[k]{k} \to 1$ (since $\frac{\ln k}{k} \to 0$), $\sqrt[k]{c} \to 1$ for $c > 0$, $\left(1 + \frac{x}{k}\right)^k \to e^x$.

- $\sum \left(\dfrac{k}{2k+1}\right)^k$:
- $\sum \left(1 - \dfrac{1}{k}\right)^{k^2}$:
- $\displaystyle\sum_{k=2}^{\infty} \frac{1}{(\ln k)^k}$:
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- $\sum \left(\dfrac{k}{2k+1}\right)^k$: $\sqrt[k]{a_k} = \dfrac{k}{2k+1} \to \dfrac{1}{2}$. Converges.
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- $\sum \frac{(1+\frac{1}{k})^{k^2}}{2^k}$: $\sqrt[k]{a_k} = \frac{(1+\frac{1}{k})^k}{2} \to \frac{e}{2} > 1$. Diverges.

# Root test: examples

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- $\sum \left(\frac{k}{2k+1}\right)^k$: $\sqrt[k]{a_k} = \frac{k}{2k+1} \to \frac{1}{2}$. Converges.
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- $\sum \frac{(1+\frac{1}{k})^{k^2}}{2^k}$: $\sqrt[k]{a_k} = \frac{(1+\frac{1}{k})^k}{2} \to \frac{e}{2} > 1$. Diverges.

The root test suits terms that are $k$-th powers.

# Ratio or root?

**Example**

$a_k = 2^{(-1)^k - k}$: $\quad \frac{1}{4},\ \frac{1}{2},\ \frac{1}{16},\ \frac{1}{8},\ \frac{1}{64},\ \frac{1}{32},\ \ldots$ (the powers of $\frac{1}{2}$, swapped in pairs).

- The ratios are $2, \frac{1}{8}, 2, \frac{1}{8}, \ldots$: no limit, so the ratio test does not apply.
- $\sqrt[k]{a_k} = 2^{(-1)^k/k - 1} \to \frac{1}{2} < 1$: the root test shows convergence.

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- $\sqrt[k]{a_k} = 2^{(-1)^k/k - 1} \to \frac{1}{2} < 1$: the root test shows convergence.

**Remark**

If $\frac{a_{k+1}}{a_k} \to L$, then also $\sqrt[k]{a_k} \to L$. So the root test decides every series that the ratio test decides — but ratios are often much easier to compute, for instance with factorials.

06

# Choosing a test

# A strategy

For $\sum a_k$ with $a_k \geq 0$:

1. Does $a_k \to 0$? If not, the series diverges.
2. Do you recognise it? Geometric $\sum r^k$, $p$-series $\sum \frac{1}{k^p}$, telescoping.
3. Factorials or exponentials such as $k!$, $c^k$? Try the **ratio test**.
4. Is the whole term a $k$-th power, $(\dots)^k$? Try the **root test**.
5. Powers, roots, quotients of polynomials in $k$? **Limit comparison** with $\frac{1}{k^p}$.
6. $a_k = f(k)$ with $f$ decreasing and an antiderivative you can find? **Integral test**.
7. Terms of both signs? Lecture 3.

# Worked examples (1)

- $\sum \dfrac{k+5}{k^3+2k}$
- $\sum k^2 e^{-k}$
- $\sum \left(\dfrac{3k}{3k+1}\right)^k$

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  Ratio test: $\left(\frac{k+1}{k}\right)^2 e^{-1} \to \frac{1}{e} < 1$. Converges.
- $\displaystyle\sum \left(\frac{3k}{3k+1}\right)^k$
  The root test gives $L = 1$: no information. But $a_k = \left(1 - \frac{1}{3k+1}\right)^k \to e^{-1/3} \neq 0$. Diverges.

# Worked examples (2)

- $\sum \dfrac{1}{k^{1+1/k}}$
- $\sum \ln\!\left(1+\dfrac{1}{k^2}\right)$
- $\displaystyle\sum_{k=2}^{\infty} \dfrac{1}{k\sqrt{\ln k}}$

# Worked examples (2)

- $\sum \dfrac{1}{k^{1+1/k}}$
  The exponent is above 1, but tends to 1. Limit comparison with $\frac{1}{k}$: $\frac{1}{\sqrt[k]{k}} \to 1$. Diverges.
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  Limit comparison with $\frac{1}{k^2}$: $\frac{\ln(1+x)}{x} \to 1$ as $x \to 0$. Converges.
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- $\displaystyle\sum_{k=2}^{\infty} \frac{1}{k\sqrt{\ln k}}$
  Integral test: $\displaystyle\int_2^t \frac{\mathrm{d}x}{x\sqrt{\ln x}} = 2\sqrt{\ln t} - 2\sqrt{\ln 2} \to \infty$. Diverges.

07

# Summary and practice

# Summary

For series with $a_k \geq 0$: converges $\iff$ partial sums bounded.

| Test | Use it when | Conclusion |
|---|---|---|
| $n$-th term | always, first | $a_k \not\to 0 \Rightarrow$ diverges |
| Integral | $a_k = f(k)$, $f \geq 0$ decreasing | $\sum a_k$ and $\int_1^\infty f$ behave alike |
| Comparison | $0 \leq a_k \leq b_k$ | $\sum b_k < \infty \Rightarrow \sum a_k < \infty$ |
| Limit comparison | $\frac{a_k}{b_k} \to L \in (0, \infty)$ | both converge or both diverge |
| Ratio | factorials, exponentials | $L < 1$ converges, $L > 1$ diverges |
| Root | $k$-th powers | $L < 1$ converges, $L > 1$ diverges |

Benchmarks: $\sum r^k$ converges iff $|r| < 1$; $\sum \frac{1}{k^p}$ converges iff $p > 1$.

# Practice

Decide whether each series converges, and name the test you use.

1. $\displaystyle\sum \frac{k}{k^3+1}$
2. $\displaystyle\sum \frac{1}{2k-1}$
3. $\displaystyle\sum \frac{k^k}{k!}$
4. $\displaystyle\sum \frac{(k!)^2}{(2k)!}$
5. $\displaystyle\sum \left(1+\frac{1}{k}\right)^{-k^2}$
6. $\displaystyle\sum \frac{\ln k}{k^{3/2}}$
7. $\displaystyle\sum \frac{k^{1000}}{1.001^k}$

# Practice, continued

8. For which $p$ does $\displaystyle\sum_{k=2}^{\infty} \frac{1}{k(\ln k)^p}$ converge?
9. Use the remainder estimate to find $\sum_{k=1}^{\infty} \frac{1}{k^3}$ to within $0.005$. How many terms do you need?
10. Show that if $a_k \geq 0$ and $\sum a_k$ converges, then $\sum a_k^2$ converges. Is the converse true?
11. Show that the comparison test is false without the condition $a_k \geq 0$: find $a_k \leq b_k$ with $\sum b_k$ convergent and $\sum a_k$ divergent.

# Answers

1. Converges: limit comparison with $\frac{1}{k^2}$.
2. Diverges: limit comparison with $\frac{1}{k}$, $L = \frac{1}{2}$.
3. Diverges: ratio $\left(1+\frac{1}{k}\right)^k \to e > 1$.
4. Converges: ratio $\frac{(k+1)^2}{(2k+2)(2k+1)} \to \frac{1}{4}$.
5. Converges: root $\left(1+\frac{1}{k}\right)^{-k} \to \frac{1}{e}$.
6. Converges: limit comparison with $\frac{1}{k^{5/4}}$, $L = \lim \frac{\ln k}{k^{1/4}} = 0$.
7. Converges: ratio $\left(1+\frac{1}{k}\right)^{1000}/1.001 \to \frac{1}{1.001}$.

# Answers, continued

8. $p > 1$ (integral test with $u = \ln x$).
9. $R_n \le \int_n^\infty x^{-3}\,\mathrm{d}x = \frac{1}{2n^2} \le 0.005$ for $n = 10$: $1.20166 \le \sum \frac{1}{k^3} \le 1.20254$.
10. $a_k \to 0$, so $a_k \le 1$ and $a_k^2 \le a_k$ for $k \ge N$. Converse false: $a_k = \frac{1}{k}$.
11. $a_k = -1,\ b_k = 0$.

# Next time

**Lecture 03 — Absolute and conditional convergence** (Unit 3)

- Series with terms of both signs; absolute convergence
- Alternating series: $\sum \frac{(-1)^{k+1}}{k}$ converges (to $\ln 2$), although $\sum \frac{1}{k}$ does not
- Rearrangements: reordering the terms can change the sum

Reading: Strang & Herman, *Calculus, Volume 2*, Section 5.5.

Questions: the module channel on Discord, or office hours, Monday–Friday, 10:00–17:00 (Baku time, UTC+4), ali.khudiyev@wosm.academy.